Agent and Patient Categories in English-Speaking Children and Homesigners

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Background: Event Participants

- Languages encode events with functions (verbs) and arguments (nouns).
  - Cross-linguistic comparison reveals languages distinguish between **agent** and **patient** roles [1, 2].
- **Are the concepts of agent and patient…**
  - …absorbed from the **linguistic input (external)?**
  - …or imposed on language by **the learner (internal)**?

How do you test this question? Look for access and deployment of concepts in individuals with no language model.

Homesigners: deaf individuals who do not have access to an external language and instead developed their own language (signs and gestures specific to the individual) [3].

Research Goals

1. Test whether the concepts of agent and patient can be deployed in an **explicit task by a population with considerable language experience but little to no test experience (4- to 5-year-olds).**
2. Determine whether the concepts of agent and patient are **accessible in the absence of a language model (homesigners*).**
   * Who also have little to no formal schooling.

Participants

	Exp 1: Adults	Exp 2: Children	Exp 3a: Homesigners* (2024)	Exp 3b: Homesigners* (2025) *slower stimuli*
N (after exclusions)	90	20	7	7
Excluded (failed training)	75 (attention check)	5	3 (in patient condition)	No exclusions
Demographics	M_{age} = 34 45 women	M_{age} = 4;11 12 girls	M_{age} = 46 4 women	M_{age} = 45 3 women

* six homesigners participated in both Exp 3a and Exp 3b.

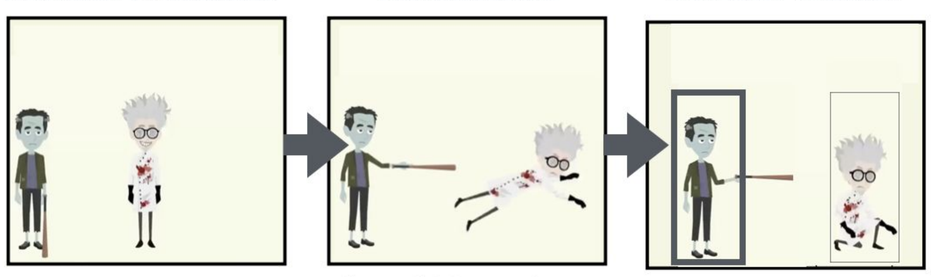
Methods and Materials

Imitation Task: Series of non-linguistic animated videos. Through scaffolding participants were trained to **select the agent or patient** and tested on their ability to generalize to **new actions** and a **new class of actions**.

Video Structure (Contact)

Character Presentation — *Contact Action* — *Character Selection*

Video Structure (Caused-Motion)



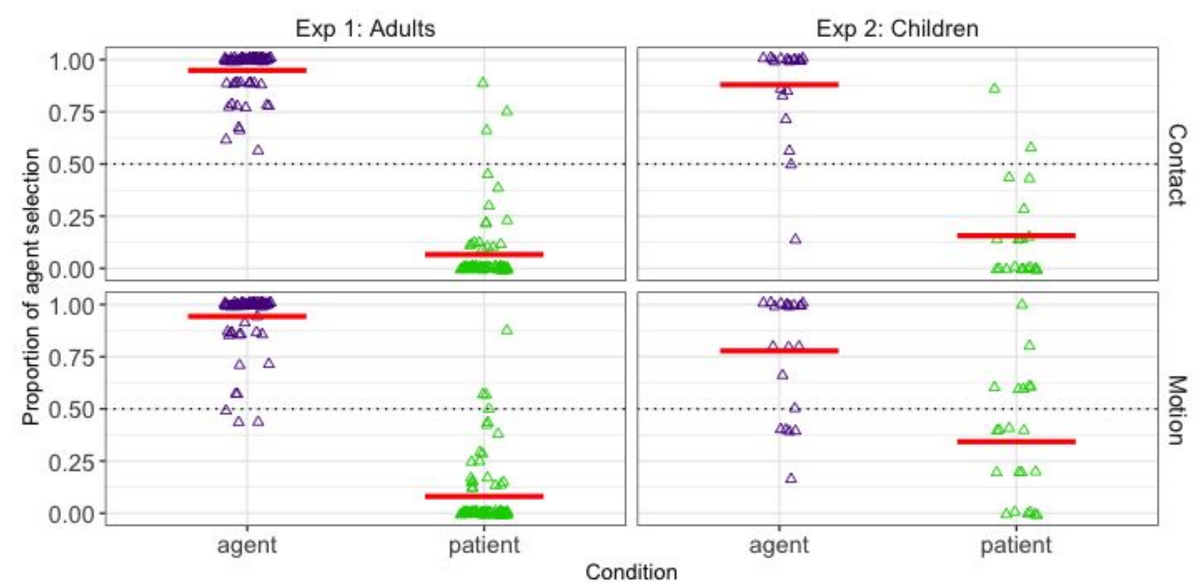
Character Presentation — *Caused-Motion Action* — *Character Selection*

Two blocks (agent selection, patient selection), order counterbalanced

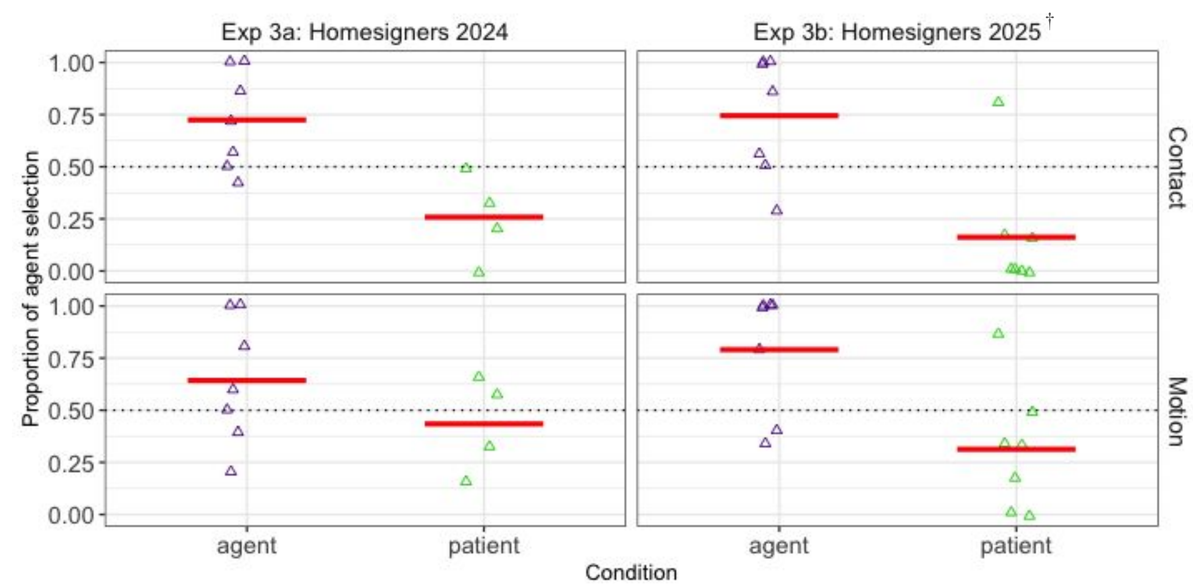
Training Phase (feedback) Contact Actions Only			Test Phase (no feedback) Contact & Caused-Motion
Demonstration phase (4 trials) *Experimenter selects and participant watches*	**Follow-me phase (6 trials)** *Experimenter selects and participant copies*	**Feedback phase (12 trials)** *Participant selects and receives feedback*	**Test phase (16 trials)** *Participant selects and receives no feedback*

Results

Experiments 1 and 2: Adults and Children



Experiments 3a and 3b: Homesigners



Analyses:

- Reliable categorical distinction between agents and patients in test phase of the categorization task.
  - Adults (Exp 1): $\boldsymbol{\beta} = 5.92$, $SE = .2$, $p < .001$
  - Children (Exp 2): $\boldsymbol{\beta} = 4.03$, $SE = .41$, $p < .001$
  - Homesigners (Exp 3a): $\boldsymbol{\beta} = 1.75$, $SE = .51$, $p < .001$
  - Homesigners (Exp 3b): $\boldsymbol{\beta} = 3.01$, $SE = .48$, $p < .001$
- Systematic extension to caused-motion actions in all populations.
  - Adults (Exp 1): $\boldsymbol{\beta} = 4.13$, $SE = .15$, $p < .001$
  - Children (Exp 2): $\boldsymbol{\beta} = 1.07$, $SE = .25$, $p < .001$; but performance in contact actions was better than in caused-motion actions ($\boldsymbol{\beta} = .06$, $SE = .24$, $p < .001$)
  - Homesigners (Exp 3a): Wilcoxon signed rank test ($V = 40.5$, $p = .2$)
  - Homesigners (Exp 3b): Wilcoxon signed rank test ($V = 79$, $p = .02$)

†One homesigner misunderstood the task and used the category they learned in the first block (patient) during the second block, resulting in 0/16 correct. When retested on the second block only with clearer instructions, they switched categories producing 16/16 correct responses. We included this 2nd session in our analysis.

Discussion

- Our task evokes agent and patient concepts in 4 to 5-year-old children.
- Homesigners can use agent and patient concepts and generalize them to novel actions and novel action class.
- That suggests that concepts of agent and patients are coming from **inside out.**

Future Directions:

- Reverse the direction of the generalization (caused-motion →contact).
- Test generalization to other action classes.
- Adapting paradigm to run with preverbal infants (looking time when trained on Agent selection and shown Patient selection or vice versa).

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